

KYUNGMIN LEE

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SUMMARY

Machine learning engineer at Rep AI with one year building production computer-vision systems, from fine-tuning transformers to shipping real-time pose feedback. M.S. Computer Science candidate at Georgia Tech.

EXPERIENCE

Machine Learning Engineer

Aug 2025 – Present

Rep AI

San Diego, CA

- Raised posture-classification accuracy from **68% to 82%** on an in-house exercise motion model by auditing pose-training data, fixing mislabeled samples, and rebuilding preprocessing
- Moved video processing and model training onto AWS EC2/S3, benchmarking c6i and g4dn instances to cut iteration time from **6 hours to under 3**
- Built a lightweight hip-alignment scoring system from joint-angle slopes rather than a second neural model, normalizing for camera perspective and filtering landmark jitter to deliver real-time squat-form feedback to **200+ users**

Search Engine Optimization & Marketing Intern

Jun 2024 – Aug 2024

Flows Tasty Treats

Newark, NJ

- Audited site metadata, backlinks, and page structure, raising the Google Lighthouse SEO score from **65 to 82**
- Benchmarked pricing and keywords across 20+ competing food businesses on Amazon and Walmart, informing a sales strategy that lifted average per-event revenue from **\$45 to \$175**

PROJECTS

Food Chart | PyTorch, DeBERTa, Flask, React Native, Docker, AWS | [Code](#) | [Demo](#)

May 2025 – Aug 2025

- Replaced LLM API calls with a fine-tuned DeBERTa regressor for ingredient weight prediction, cutting response time from **3s to 1s** and external API cost to zero, with fallback to a larger model only on low-confidence cases
- Built a **26K-row** training set from USDA data and scraped recipes to normalize measurements such as “a bunch of parsley”, reaching 76% accuracy

Outfit Finder | Python, Flask, Claude API, Playwright, asyncio | [Code](#)

Jul 2026

- Cut end-to-end runtime from **653s to 60s** by moving orchestration out of the LLM into deterministic Python, reducing the model to stateless JSON transforms behind explicit timeouts and fail-soft stages
- Ranked products by parsing 1–3 star reviews for defect signals such as shrinkage and stitching failures, surfacing quality information keyword search cannot reach

Muscle 3D | Next.js, three.js, MediaPipe, Electron, Capacitor, Supabase | [Code](#)

2026

- Shipped a 3D anatomy reference to **200+ users** as separate desktop and mobile builds from one Next.js codebase spanning web, macOS, and iOS
- Reduced asset size by over **70%** by stripping unused meshes in Blender and tuning caching, and avoided cloud video storage by serving community-submitted YouTube links

Posture Coach | Swift, Apple Vision, MediaPipe, Next.js | [Code](#)

2026

- Built one posture-coaching product in two implementations: MediaPipe landmarking on the web, and Apple Vision in native Swift for macOS and iOS from a shared codebase

EDUCATION

Georgia Institute of Technology

Expected May 2028

Master of Science in Computer Science

Rutgers University, New Brunswick

May 2025

Bachelor of Arts in Computer Science — GPA: 3.68/4.00

TECHNICAL SKILLS

ML/AI: PyTorch, Hugging Face Transformers, DeBERTa, fine-tuning, MediaPipe, LLM orchestration, TensorFlow

Languages: Python, JavaScript/TypeScript, SQL, Swift, Java, C/C++

Web/Mobile: React, React Native, Next.js, Node.js, Flask, Expo, Capacitor, three.js

Infrastructure: AWS (EC2, S3, SageMaker), Docker, Supabase, Vercel, Git